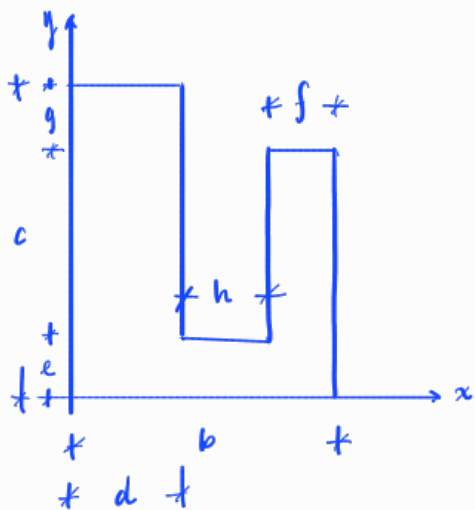
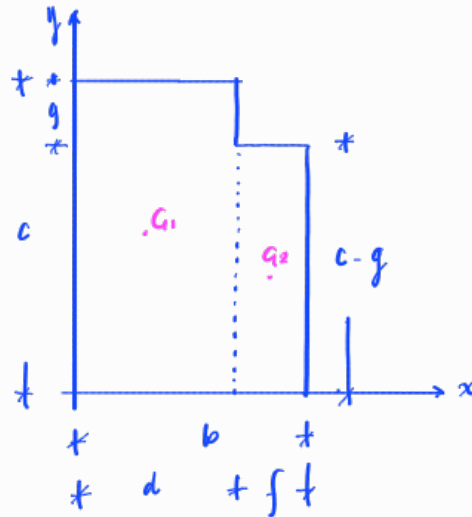




$$\begin{aligned} b &= 10a \\ c &= 14a \\ d &= 6a \\ f &= g = 4a \end{aligned}$$

$$h = b - (f + d) = 10a - (4a + 6a) = 0$$



$$G_1 \left(\frac{d}{2}; \frac{c}{2} \right) = (3a; 7a)$$

$$G_2 \left(d + \frac{f}{2}; \frac{c-g}{2} \right) = (8a; 5a)$$

$$A_{TOT} = A_1 + A_2 = 124a^2$$

$$A_1 = c \cdot d = 14a \cdot 6a = 84a^2$$

$$A_2 = (c-g) \cdot f = 10a \cdot 4a = 40a^2$$

Momenti statici:

$$\left. \begin{aligned} S_{X_1} &= A_1 \cdot y_{G_1} = 84a^2 \cdot 7a = 588a^3 \\ S_{X_2} &= A_2 \cdot y_{G_2} = 40a^2 \cdot 5a = 200a^3 \end{aligned} \right\} S_X = S_{X_1} + S_{X_2} = 788a^3$$

$$\left. \begin{aligned} S_{Y_1} &= A_1 \cdot x_{G_1} = 84a^2 \cdot 3a = 252a^3 \\ S_{Y_2} &= A_2 \cdot x_{G_2} = 40a^2 \cdot 8a = 320a^3 \end{aligned} \right\} S_Y = S_{Y_1} + S_{Y_2} = 572a^3$$

$$x_G = \frac{S_Y}{A_{TOT}} = 4,61a$$

$$y_G = \frac{S_X}{A_{TOT}} = 6,35a$$

Momenti di inerzia

$$J_{X_1} = \frac{1}{3} d c^3 = \frac{1}{3} 6a (14a)^3 = 5438a^4$$

$$J_{X_2} = \frac{1}{3} f (c-g)^3 = \frac{1}{3} 4a (10a)^3 = 1333,33a^4$$

$$J_{Y_1} = \frac{1}{3} d^3 c = \frac{1}{3} (6a)^3 \cdot 14a = 1008a^4$$

$$J_{Y_2} = \frac{1}{12} f^3 (c-g) + A_2 \cdot x_G^2 = 2613,33a^4$$

$$J_X = J_{X_1} + J_{X_2} = 6821,33a^4$$

$$J_Y = J_{Y_1} + J_{Y_2} = 3621,33a^4$$

$$J_{X_G} = J_X - A y_G^2 = 6821,33a^4 - 124a^2 (6,35a)^2 = 1813,72a^4$$

$$J_{Y_G} = J_Y - A x_G^2 = 3621,33a^4 - 124a^2 (4,61a)^2 = 982,76a^4$$

